PEER REVIEW BY SHARED SERVICES CANADA OF THE ADVANCED ANALYTICS TRIAGE FOR VISITOR RECORD APPLICATIONS

EXECUTIVE SUMMARY

Immigration, Refugees and Citizenship Canada
Immigration, Réfugiés et Citoyenneté Canada
Canada

Executive Summary

The peer review report assesses an Artificial Intelligence application developed by Immigration, Refugees and Citizenship Canada (IRCC) for the *Visitor Record* (VR) program. An assessment is a mandatory requirement of the Treasury Board of Canada Secretariat's (TBS) Directive on Automated Decision-Making and evaluates the IRCC's machine learning models that assist with the eligibility assessment for all visitor record applications. The IRCC application under review flags uncomplicated applications for automated eligibility approval by leveraging a series of rules developed from IRCC personnel in tandem with a custom machine learning model. The assessment finds that IRCC's solution is robust, and the necessary controls and safeguards have been sufficiently described to ensure applicant privacy is preserved throughout its lifecycle.

Approach

During the peer review process, a team of Shared Services Canada (SSC) data scientists and business analysts worked with IRCC to understand the Visitor Request process and the AI support tools. The SSC team referred to the Bronson/Millar *Peer Review for Automated Decision-Making Tools Under Canada's Directive on Automated Decision-Making – A Model Peer Review Process and Recommendations for Best Practices*, TBS *Guide on Peer Review (Draft)* and associated checklist, the TBS Directive on Automated Decision Making (Directive), to guide the assessment process.

Findings

Throughout the assessment process SSC discovered instances where provided documentation did not fully reflect the current deployment of the system, and relevant information from discussions was absent. All discovered discrepancies were reported to IRCC for investigation. IRCC has confirmed their intent to resolve all discrepancies.

The review demonstrates that the IRCC staff have developed their models consistent with TBS objectives for transparency and risk mitigation.

Recommendations

Fairness

Discrimination risk from relabeling records: Any adjustments (relabeling) made to the training data should be carefully evaluated to ensure they do not introduce bias against specific groups of applicants. If the data on infractions and the reversal of decisions from

approved to *refused* is not representative of all groups, there is a risk that the model could learn to discriminate based on those biases. IRCC recognizes there is always inherent risk of introducing bias when relabeling training data. As future iterations of the model are developed, IRCC commits to continue monitoring the model to ensure no specific groups of applicants experience bias.

Discrimination risk from external measurements and/or adverse events: In future training iterations, it is recommended to clearly distinguish records that were approved or refused based on permanent or temporary adverse events or external such as a changing health situation, political/governmental factors or other adverse events to avoid introducing permanent classifications or other forms of discrimination. As the final decision is still overseen by an officer, *the overall risk remains low* but should be clearly reported. IRCC will continue its practice of gathering information about changing country conditions to ensure that decisions made by models and humans incorporate recent developments. The risk of bias will be mitigated by vetting any new rules and by performing Gender-Based Analysis Plus of the triage results.

Overreliance

Officer trust in the triage process may bias decisions: There is a risk that officers may become overly reliant on the triage system, potentially impacting their decision-making on admissibility. Officers may not navigate the multiple tabs and fields to appreciate the full assessment of the model. It is important to emphasize that officers must maintain a critical approach when reviewing applications, ensuring they do not blindly trust the automated outputs. Additionally, there is a need to closely monitor decisions related to adverse events to ensure they are thoroughly reviewed. IRCC recognizes that automation bias is an inherent risk in all automation models. To account for this, the VR Model's QA plan includes a "blind" QA component to mitigate the risk of automation bias on decision-makers.

Model

Model Validation through Cross-Validation would increase performance: To enhance the model's generalization capabilities, it is recommended to implement cross-validation techniques during training. This approach will help verify the model's performance on unseen data and ensure that it is robust against overfitting, thereby improving its reliability in real-world applications.

The IRCC Advanced Analytics Solution Centre (A2SC) decided that for future models, cross-validation steps will be used for model training in order to prove robustness of the model. However, the current model building procedures include a testing step which uses 30% of unseen test data to ensure absence of overfitting.

Privacy

Privacy Data Management and Security will continue to be ongoing Risks: As reported in the provided documentation and associated annexes, the data used for training the models contains sensitive personal information and is managed under strict security protocols. Despite being hosted in a protected cloud environment, interfacing with other systems and transferring data could pose potential risks. Ongoing vigilance and robust security measures are essential to mitigate risks associated with data inconsistencies or inaccuracy. IRCC will maintain its vigilance with respect to the internal dissemination of personal information.

Data

Exploiting free text fields may bring value: SSC recommends conducting periodic reviews of text mining results in the free text fields to verify data accuracy and control. The business has asked the A2SC to perform text mining analysis of these fields. A2SC is in the process of completing this analysis and will continue to do so periodically.

General Recommendations

Additional information discovered throughout the peer review process should be integrated into the original documentation.